

# Suraj Krishna Banavalikar

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🌐 [github.com/surajkb2005](https://github.com/surajkb2005) | [linkedin.com/in/suraj-banavalikar-7043202a5](https://www.linkedin.com/in/suraj-banavalikar-7043202a5) | 🌐 Portfolio

## Skills

**Programming Languages:** C/C++, Java, Python, JavaScript

**Core Computer Science:** Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks

**Databases:** PostgreSQL, MySQL, MongoDB

**Technologies & Tools:** Git, GitHub, Linux, Docker

**Frontend Development:** HTML, CSS, React.js

**Backend & Systems:** Node.js, REST APIs

## Education

**Ramaiah Institute of Technology (MSRIT) | Bengaluru**

B.E. in Computer Science and Engineering

Oct 2023 - Present

**CGPA: 9.21/10**

**St. Joseph's PU College | Karwar**

Course in PCMC

Aug 2021 - March 2023

**Grade: 96.16%**

## Project Work

**Campus Helpdesk – AI Powered Campus Helpdesk System (2025) | MERN Stack, Python, Pinecone, JavaScript**

- Built an NLP-powered query system (vector search + hybrid re-ranking with Pinecone) improving answer accuracy and reducing manual query handling by 40%.
- Optimized performance with MongoDB connection pooling + async job queues (20 jobs/batch, retry with backoff), reducing API blocking and improving throughput under concurrent load.
- Architected a scalable Node.js backend using PM2 clustering (10 workers) with load balancing, supporting 50–100 concurrent users and stable low-latency responses.
- **Source Code:** [surajkb2005.github.io/Resume/pages/campus\\_helpdesk.html](https://surajkb2005.github.io/Resume/pages/campus_helpdesk.html)

**StellarRoutes (2025) | Python, Leaflet.js, NASA APIs, Redis**

- Designed and implemented a GPS-denied navigation system using inertial navigation (IMU-based dead reckoning).
- Processed accelerometer and gyroscope data to estimate position in high-interference environments, such as solar storms, to calculate position without satellite dependency.
- Integrated NASA's NOVA API for live kp index values and proposed future integration with ISRO's Aditya-L1 real-time telemetry.
- **Source Code:** [github.com/varun99015/StellarRoute](https://github.com/varun99015/StellarRoute)

**Autonomous Obstacle Avoidance Rover | Arduino UNO, C Programming, Ultrasonic Sensors**

- Built an embedded autonomous navigation system using real-time sensor feedback and decision algorithms.
- Implemented closed-loop control for obstacle avoidance with dynamic path planning (turn, move, reverse).
- Analyzed limitations such as sensor noise, drift, and scalability, proposing system-level improvements.
- **Source Code:** [github.com/surajkb2005/Obstacles\\_Avoidance\\_Land\\_Rover](https://github.com/surajkb2005/Obstacles_Avoidance_Land_Rover)

## Work Experience

**RIT S.T.A.R.D.U.S.T**

May 2025 - Present

**Software Engineer – Avionics Member**

- Contributed to early-stage development of software modules for autonomous drone systems.
- Assisted in simulation and testing of navigation and control logic.

## Awards and Certificates

- Secured 2nd Place at wHACKiest 2025 (24-hour Intra-college level hackathon) among 100+ teams. Built the Stellar-Route project as part of a team, contributing as a backend engineer.

## Relevant Interests

- Scalable Backend Systems, Distributed Architectures and Navigation Algorithms.
- Autonomous Systems, Space Technology, and Hardware Prototyping (Arduino)